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School of Sciences
(Department of Mathematics)

APPLIED CALCULUS LAB
(26SOSP103)

MATLAB LABORATORY MANUAL



B.Tech. Programme (first sem.)
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Name:. . . . . .

Roll No:. . . . . .

Group:. . . . . .

Instructor:. . . . . .

S. No.	List of Experiments	Date	Signature
1	Introduction to MATLAB: Basic commands, symbolic variables, arithmetic operations, and plotting of standard functions		
2	Graphing functions and derivatives		
3	Successive differentiation and Leibnitz's Theorem of algebraic, trigonometric, exponential, and logarithmic functions		
4	Taylor and Maclaurin series expansion of one-variable functions and approximation of function values		
5	Graphical verification of Rolle's Theorem and Lagrange's Mean Value Theorem		
6	Homogenous functions, Euler’s theorem and Multivariable Taylor series		
7	Determination and Visualization of Maxima and Minima of Functions of Two Variables		
8	Gradient Descent for Function Minimization		
9	Newton’s method for Function Minimization		
10	Single and Multiple Linear Regression using Gradient Descent		

Experiment 1A : Introduction to MATLAB: Basic commands, symbolic variables, arithmetic operations, and plotting of standard functions

1. Write an algorithm for Introduction to MATLAB (Basic commands, symbolic variables, arithmetic operations, and plotting of standard functions).

Algorithm for Introduction to MATLAB (Basic commands, symbolic variables, arithmetic operations, and plotting of standard functions): To familiarize students with the MATLAB environment, variables, basic commands, arithmetic operations, symbolic variables, built-in mathematical functions, and graphical visualization of standard mathematical functions.

Input: Numerical values, symbolic variables, and mathematical expressions.

Output: Numerical results, symbolic computations, and graphical plots.

Step 1: Launch MATLAB and open a new script file or notebook.

Step 2: Explore the development environment, including the Command Window/Console, Workspace or Variable Explorer, Current Folder, Editor, and Command History.

Step 3: Create variables and assign numerical values.

Step 4: Perform arithmetic operations such as addition, subtraction, multiplication, division, modulus, and exponentiation.

Step 5: Evaluate built-in mathematical functions such as square root, exponential, logarithmic, and trigonometric functions.

Step 6: Define symbolic variables and create symbolic mathematical expressions.

Step 7: Perform symbolic computations such as simplification, expansion, and factorization.

Step 8: Generate a suitable range of values for the independent variable.

Step 9: Plot a standard mathematical function using appropriate plotting commands.

Step 10: Add a title, axis labels, legend (if required), and grid to the graph.

Step 11: Display the numerical results, symbolic results, and graphical output.

Step 12: Save the program and terminate the execution.

2. Create two variables a =15 and b = 5. Perform the following operations:

- Addition
- Subtraction
- Multiplication
- Division
- Exponentiation
- Modulus

3. Evaluate the following mathematical expressions: $\sqrt{64}$, e^2, $sin\left(\frac{\pi}{6}\right)$, $cos\left(\frac{\pi}{3}\right)$, $log(100)$.

4. Define the symbolic expression x^2+5x+6. Using MATLAB, Factorize the given expression.

5. Plot the following functions over the interval $0 \leq x \leq 2\pi$.

- $y = sin\ x$
- $y = cos\ x$

Include the graph title, axis labels, and grid.

6. A company's monthly profit is modeled by $P(x) = x^2+4x+5$, where x represents the number of months. Using MATLAB,

- Define the function.
- Evaluate the profit for $x = 5$
- Plot the function for $0 \leq x \leq 10$.

Experiment 1B : Introduction to MATLAB: Decision Making and Looping Statements in MATLAB

1. Write an algorithm for Introduction to MATLAB (Decision Making and Looping Statements in MATLAB).

Algorithm for Introduction to MATLAB (Decision Making and Looping Statements in MATLAB): To familiarize students with conditional statements and iterative programming using MATLAB.

Input: Numerical values and logical conditions.

Output: Results obtained using conditional statements and looping constructs.

Step 1: Launch MATLAB and create a new script file.

Step 2: Define the required variables.

Step 3: Read or assign input values.

Step 4: Implement conditional statements (if, if-else, nested if, or switch-case/match-case) to evaluate logical conditions.

Step 5: Implement a for loop to execute statements for a fixed number of iterations.

Step 6: Implement a while loop to execute statements until a specified condition becomes false.

Step 7: Use break and continue statements to control loop execution where required.

Step 8: Display the output.

Step 9: Save the program and terminate the execution.

2. Write a MATLAB program to determine whether a given number is positive, negative, or zero.

3. Write a MATLAB program to determine whether a given year is a leap year.

4. Write a MATLAB program to find the largest among three numbers using the **if-else** statement.

5. Using a **for** loop, compute the sum of the first 20 natural numbers.

6. Using a **while** loop, generate the multiplication table of a given number up to 10.

7. Using a **for** loop, generate and display the squares and cubes of the numbers from **1 to 10** in tabular form.

Experiment 2 : Graphing functions and derivatives

1. Write an algorithm for Graphing functions and derivatives.

Algorithm for Graphing functions and derivatives: To visualize the relationship between a function and its derivative by plotting both on the same coordinate system using MATLAB and to interpret the geometric meaning of the derivative.

Input: A differentiable function $f(x)$

Output: Graphs of the function and its derivative, and graphical interpretation of their relationship.

Step 1: Launch MATLAB and create a new script file.

Step 2: Define the independent variable x.

Step 3: Enter the function $f(x)$.

Step 4: Compute the first derivative $f'(x)$ symbolically or numerically.

Step 5: Generate a suitable range of values for x.

Step 6: Evaluate both $f(x)$ and $f'(x)$ over the specified interval.

Step 7: Plot the function $f(x)$ and its derivative $f'(x)$ on the same graph using different colors or line styles.

Step 8: Add an appropriate title, axis labels, legend, and grid to the graph.

Step 9: Observe the relationship between the function and its derivative by identifying intervals where the function is increasing, decreasing, or has stationary points.

Step 10: Display the graphical output and interpret the results.

Step 11: Save the program and terminate the execution.

2. Using MATLAB, plot $f(x) = x^3 - 3x$ and its first derivative on the same graph. Identify the stationary points of the function.

3. Using MATLAB, plot $f(x) = \sin x$ and its first derivative on the same graph. Compare the graphs and verify that the derivative of $\sin x$ is $\cos x$.

Experiment 3: Successive Differentiation

1. Write an algorithm for Successive Differentiation.

Algorithm for Successive Differentiation : To determine the successive derivatives of a given symbolic function $f(x)$ up to the nth order using MATLAB.

Input: Symbolic function $f(x)$ order of differentiation n.

Output: Successive derivatives of $f(x)$ up to the nth order.

Step 1: Define the symbolic variable x.

Step 2: Enter the symbolic function $f(x)$.

Step 3: Read the order of differentiation n.

Step 4: for $i = 1 : n$

$d(i) = diff(f, x, i)$;

Display $d(i)$;

end

Step 5: Stop.

2. Write a program to determine the successive derivatives up to the fifth order of the function $f(x) = e^x \sin x$. Display all the derivatives and comment on the pattern observed in the successive derivatives.

3. Write a MATLAB program to determine the first, second, and third order derivatives of the displacement function $f(t) = 3t^4 - 5t^3 + 2t^2 + 6t + 1$. Hence, determine the expressions for velocity, acceleration, and jerk, and discuss their physical significance.

Experiment 4: Taylor series expansion of one-variable functions

1. Write an algorithm for Function Approximation Using Taylor Series

Algorithm for Taylor series expansion of one-variable functions: To determine the approximate value of a given function using its Taylor series expansion.

Input : Symbolic function $f(x)$, expansion point a, approximation point x_0, degree n

Output : Approximate value of $f(x_0)$

Step 1: Define the symbolic variable x.

Step 2: Enter the symbolic function $f(x)$.

Step 3: Read the values of a, x_0 and n.

Step 4: Compute the Taylor series expansion

T = taylor(f, x, 'ExpansionPoint', a, 'Order', $n+1$);

Step 5: Evaluate the Taylor polynomial at $x = x_0$

Step 6: Display the approximate value.

Step 7: Stop.

2. Write a MATLAB program to determine the Taylor series expansion (Maclaurin series about $x = 0$) of the function $f(x) = e^x$ up to the sixth-degree term. Display the polynomial, plot both the original function and its approximation over the interval $[-2, 2]$, and comment on the accuracy of the approximation.

3. The displacement of a vibrating particle is given by $f(t) = sin\, t$. Write a MATLAB program to find the Taylor series expansion of the given function about $t = 0$ up to the 7th-degree term. Use the Taylor polynomial to calculate the displacement at $t = 0.5$. Compare the approximate result with the exact value and determine the approximation error.

Experiment 5A : Graphical Verification of Rolle's Theorem

1. Write an algorithm for Graphical verification of Rolle's Theorem and Lagrange's Mean Value Theorem.

Algorithm of Graphical Verification of Rolle's Theorem : To determine graphically whether a given function satisfies Rolle's Theorem on the interval $[a, b]$ using MATLAB.

Input : Symbolic function $f(x)$, interval $[a, b]$.

Output : Graph of the function and the value(s) of c

Step 1: Define the symbolic variable x.

Step 2: Enter the symbolic function $f(x)$.

Step 3: Specify the interval $[a, b]$

Step 4: Verify that

(i) $f(x)$ is continuous on $[a, b]$,

(ii) $f(x)$ is differentiable on (a, b),

(iii) $f(a) = f(b)$

Step 5: Compute the first derivative D = diff(f, x).

Step 6: Solve D = 0 to obtain the value(s) of c .

Step 7: Plot the graph of $f(x)$ over the interval $[a, b]$.

Step 8: Mark the end points and the point(s) c on the graph.

Step 9: Display the graph and verify that the tangent at c is parallel to the x-axis.

Step 10: Stop.

2. The height of a road above the ground is modeled by $h(x)=x^3-6x^2+9x+1$, where x is the horizontal distance measured in kilometres. Using the MATLAB Symbolic Math Toolbox, verify Rolle's Theorem over the interval [0, 3]. Compute the point(s) at which the road has a horizontal tangent, and verify the theorem graphically by plotting the elevation curve.

Experiment 5B : Graphical verification of Lagrange's Mean Value Theorem

1. Write an algorithm for Graphical verification of Lagrange's Mean Value Theorem.

Algorithm of Graphical Verification of Rolle's Theorem : To determine graphically whether a given function satisfies Lagrange's Mean Value Theorem on the interval [a, b] using MATLAB.

Input : Symbolic function f(x), interval [a, b]

Output : Graph of the function and the value(s) of c

Step 1: Define the symbolic variable x.

Step 2: Enter the symbolic function f(x).

Step 3: Specify the interval [a, b].

Step 4: Verify that

(i) f(x) is continuous on [a, b],

(ii) f(x) is differentiable on (a, b).

Step 5: Compute the average slope m = (f(b) - f(a))/(b - a).

Step 6: Compute the first derivative D = diff(f, x).

Step 7: Solve D = m to obtain the value(s) of c.

Step 8: Plot the graph of f(x) over the interval [a, b].

Step 9: Draw the secant line joining (a, f(a)) and (b, f(b)).

Step 10: Mark the point(s) c where the tangent is parallel to the secant line.

Step 11: Display the graph and verify the theorem graphically.

Step 12: Stop.

2. The motion of a vehicle is described by the displacement function $f(t) = t^3 - 3t^2 + 2t + 5$. Using MATLAB, verify Lagrange's Mean Value Theorem on the interval [0, 2]. Compute the average rate of change of displacement over the interval and determine the corresponding instant where the instantaneous rate of change is the same. Illustrate the result graphically.

Experiment 6 : Homogeneous Functions, Euler's Theorem, and Multivariable Taylor Series

1. Write an algorithm for Homogeneous Functions, Euler's Theorem, and Multivariable Taylor Series.

Algorithm of Homogeneous Functions, Euler's Theorem, and Multivariable Taylor Series : To identify homogeneous functions, verify Euler's theorem, and compute the Taylor series expansion of multivariable functions using MATLAB.

Input: A multivariable function $f(x,y)$ and the point of expansion (a,b).

Output: Identification of the homogeneous function, verification of Euler's theorem, and Taylor series approximation.

Part A: Verification of Euler's Theorem

Step 1: Launch MATLAB and create a new script file.

Step 2: Define the symbolic variables x and y.

Step 3: Enter the given function $f(x,y)$.

Step 4: Determine whether the function is homogeneous and identify its degree nnn.

Step 5: Compute the first-order partial derivatives: f_x *and* f_y

Step 6: Compute $L = x f_x + y f_y$.

Step 7: Compute $R = n f(x,y)$

Step 8: Compare L and R. If $L = R$, conclude that Euler's theorem is verified.

Part B: Multivariable Taylor Series

Step 9: Enter the point (a,b) about which the Taylor series is to be expanded.

Step 10: Compute the required first- and second-order partial derivatives of $f(x,y)$.

Step 11: Construct the Taylor series expansion of the function about the point (a,b).

Step 12: Display the Taylor polynomial obtained.

Step 13: Save the program and terminate the execution.

2. Using MATLAB, verify Euler's theorem for $f(x, y) = x^3 + 3x^2y + 3xy^2 + y^3$.

3. Using MATLAB, obtain the second-order Taylor series expansion of $f(x,y) = e^{x+y}$ about the point (0,0).

Experiment 7 : Determination and Visualization of Maxima and Minima of Functions of Two Variables

1. Write an algorithm for Determination and Visualization of Maxima and Minima of Functions of Two Variables

Algorithm for Determination and Visualization of Maxima and Minima of Functions of Two Variables: To determine and visualize the maxima and minima of functions of two variables using MATLAB by computing critical points, applying the second derivative test, and plotting the corresponding surface.

Input: A function $f(x,y)$.

Output: Critical points, nature of the critical points, and graphical visualization of the function.

Step 1: Launch MATLAB and create a new script file.

Step 2: Define the symbolic variables x and y.

Step 3: Enter the given function $f(x,y)$.

Step 4: Compute the first-order partial derivatives f_x *and* f_y.

Step 5: Solve $f_x = 0$, $f_y = 0$ to obtain the critical points.

Step 6: Compute the second-order partial derivatives f_{xx} , f_{xy} , f_{yy}.

Step 7: Compute the Hessian determinant $D = f_{xx}\, f_{yy} - (f_{xy})^2$ at each critical point.

Step 8: Classify each critical point using the second derivative test:

- If $D > 0$ *and* $f_{xx} > 0$, the point is a **local minimum**.
- If $D > 0$ *and* $f_{xx} < 0$, the point is a **local maximum**.
- If $D < 0$, the point is a **saddle point**.
- If $D = 0$, the test is **inconclusive**.

Step 9: Generate a suitable mesh of values for x and y.

Step 10: Plot the function using a 3D surface plot and/or contour plot.

Step 11: Display the critical points, their nature, and the graphical visualization.

Step 12: Save the program and terminate the execution.

2. Using MATLAB, determine the critical points and identify the maxima or minima of $f(x,y) = x^2+y^2$. Visualize the function using a surface plot.

3. Using MATLAB determine the critical points of $f(x,y) = x\text{^}2 + y\text{^}2 - 4x - 6y$. Classify each critical point and plot the surface.

Experiment 8 : Gradient Descent for Function Minimization

1. Write an algorithm for Gradient Descent for Function Minimization

Algorithm for Gradient Descent for Function Minimization: To implement the Gradient Descent algorithm to determine the minimum of a multivariable function using MATLAB.

Input: A differentiable function $f(x,y)$ an initial point (x_0,y_0), learning rate α, tolerance ε, and maximum number of iterations.

Output: Approximate minimum point and minimum function value.

Step 1: Define the multivariable function $f(x,y)$.

Step 2: Compute the gradient components f_x, f_y

Step 3: Select an initial point (x_0, y_0).

Step 4: Choose a suitable learning rate α, tolerance ε, and maximum number of iterations.

Step 5: Evaluate the gradient at the current point.

Step 6: Update the variables using

$$x_{k+1} = x_k - \alpha f_x$$

$$y_{k+1} = y_k - \alpha f_y$$

Step 7: Compute the new function value.

Step 8: Repeat Steps 5–7 until

- the gradient norm becomes smaller than the specified tolerance, **or**
- the maximum number of iterations is reached.

Step 9: Display the approximate minimum point and the corresponding minimum function value.

Step 10: Plot the optimization path on the contour or surface plot.

Step 11: Stop.

2. Minimize $f(x,y) = x^2+2y^2$. Plot the optimization path.

3. The loss function of a simple machine learning model is $F(x,y) = x^2+y^2-6x-4y+20$. Use Gradient Descent to determine the optimal parameter values.

Experiment 9 : Newton's method for Function Minimization

1. Write an algorithm for Newton's method for Function Minimization.

Algorithm for Newton's method for Function Minimization: To implement Newton's method for determining the minimum of a multivariable function using MATLAB.

Input: A twice differentiable function $f(x,y)$, an initial point (x_0, y_0) tolerance ε\varepsilonε, and maximum number of iterations.

Output: Approximate minimum point and minimum function value.

Step 1: Launch MATLAB and create a new script file.

Step 2: Define the multivariable function $f(x,y)$.

Step 3: Compute the gradient vector $\nabla f(x,y)$

Step 4: Compute the Hessian matrix H

Step 5: Select an initial point (x_0,y_0).

Step 6: Evaluate the gradient vector and Hessian matrix at the current point.

Step 7: Update the variables using Newton's iteration $x_{k+1} = x_k - H^{-1}\nabla f(x_k)$.

Step 8: Repeat Steps 6–7 until the norm of the gradient is less than the specified tolerance or the maximum number of iterations is reached.

Step 9: Display the approximate minimum point and the corresponding minimum function value.

Step 10: Plot the function surface or contour plot and indicate the minimum point.

Step 11: Save the program and terminate the execution.

2. Using MATLAB, determine the minimum of $f(x, y) = x^2+y^2$, using Newton's method.

3. Using MATLAB, minimize $f(x,y) = x^2+2y^2-8x-12y+30$. Determine the minimum point and the minimum function value.

Experiment 10 : Single and Multiple Linear Regression using Gradient Descent

1. Write an algorithm for Single and Multiple Linear Regression using Gradient Descent

Algorithm for Single and Multiple Linear Regression using Gradient Descent : To implement single and multiple linear regression models using the Gradient Descent optimization algorithm and estimate the optimal model parameters using MATLAB.

Input: Training dataset, learning rate α, tolerance ε, and maximum number of iterations.

Output: Regression coefficients, fitted regression model, and predicted output values.

Step 1: Launch MATLAB and create a new script file.

Step 2: Enter or load the training dataset.

Step 3: Separate the independent variable(s) and the dependent variable.

Step 4: Initialize the regression parameters with suitable initial values.

Step 5: Choose the learning rate, tolerance, and maximum number of iterations.

Step 6: Compute the predicted output using the regression model.

Step 7: Evaluate the Mean Squared Error (MSE) cost function.

Step 8: Compute the gradient of the cost function with respect to each regression parameter.

Step 9: Update the regression parameters using the Gradient Descent update rule.

Step 10: Repeat Steps 6–9 until the convergence criterion is satisfied or the maximum number of iterations is reached.

Step 11: Display the optimal regression coefficients and the final cost.

Step 12: Plot the fitted regression line (for single regression) or compare the predicted and actual values (for multiple regression).

Step 13: Save the program and terminate the execution.

2. The following data represent the study hours and examination marks of five students.

Study Hours	Marks
1	40
2	50
3	60
4	70
5	80

Using MATLAB:

- Implement Single Linear Regression using Gradient Descent.

- Determine the regression equation.
- Plot the fitted regression line.
- Predict the marks of a student who studies 6 hours.